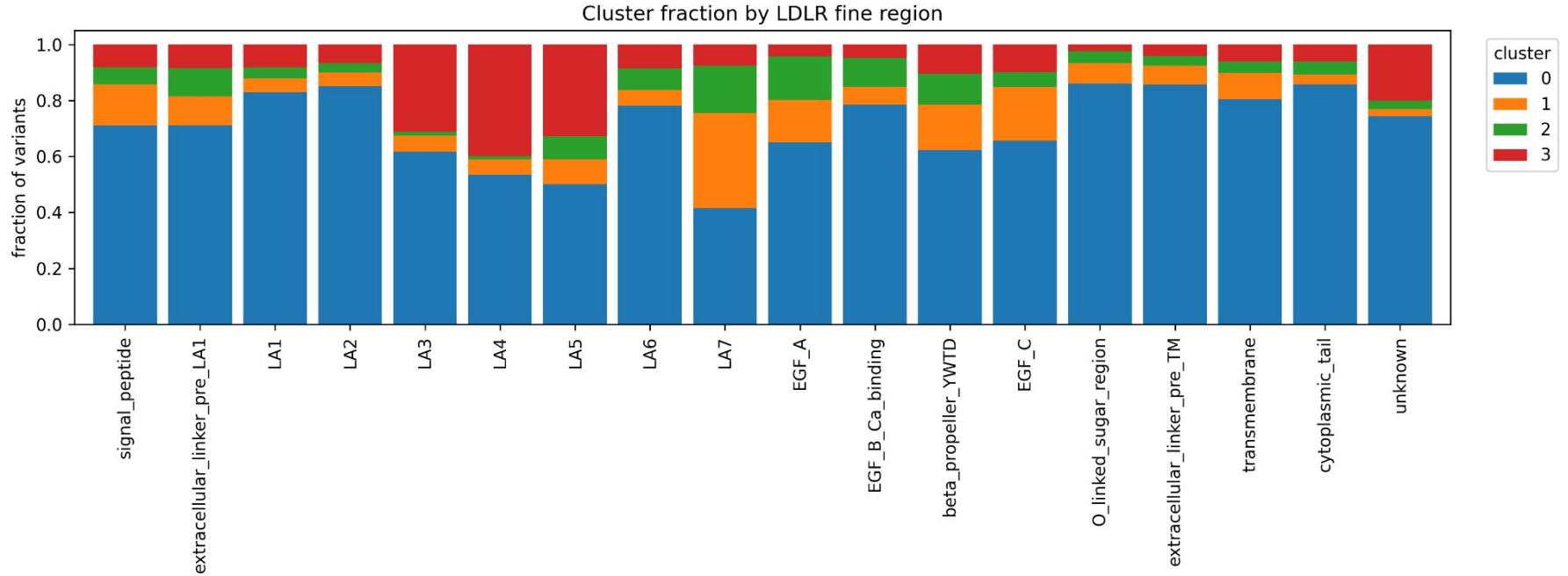


Biological Pattern Search of LDLR Variants

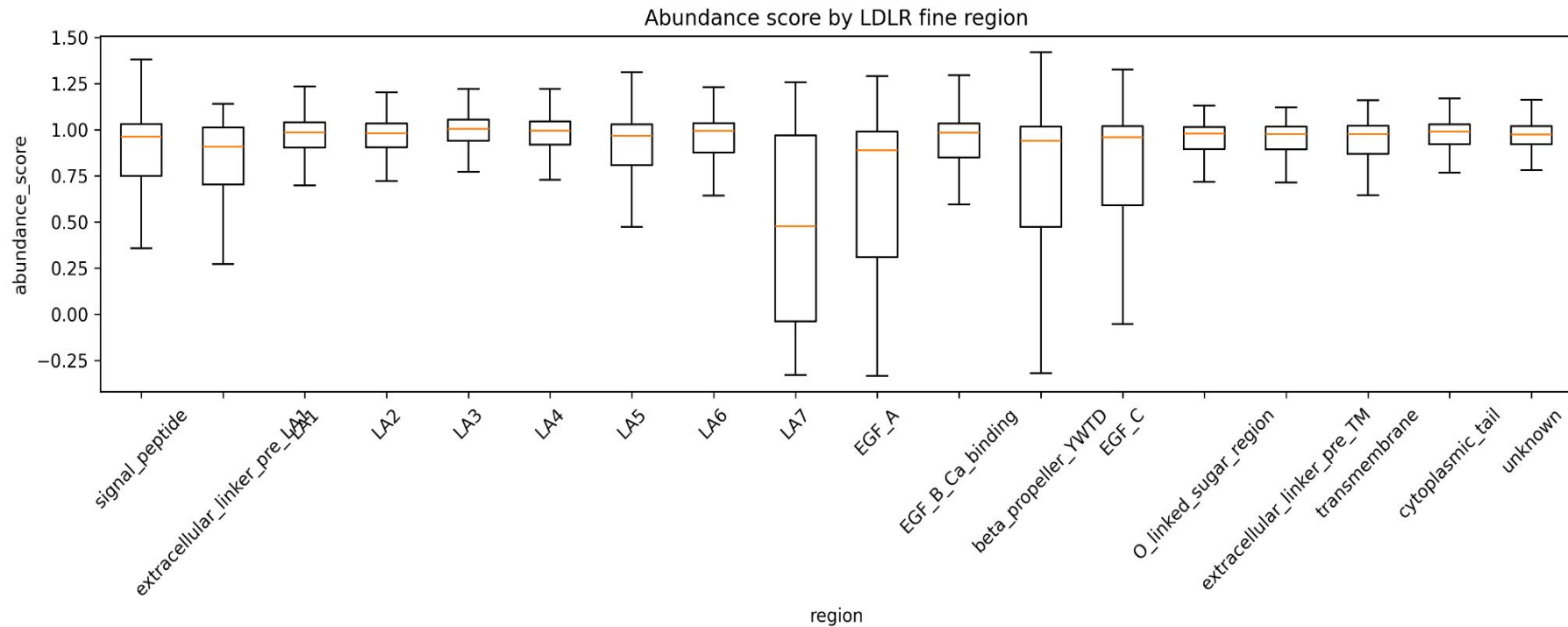
May 15

Part 1: Cluster Fraction by LDLR Fine Region

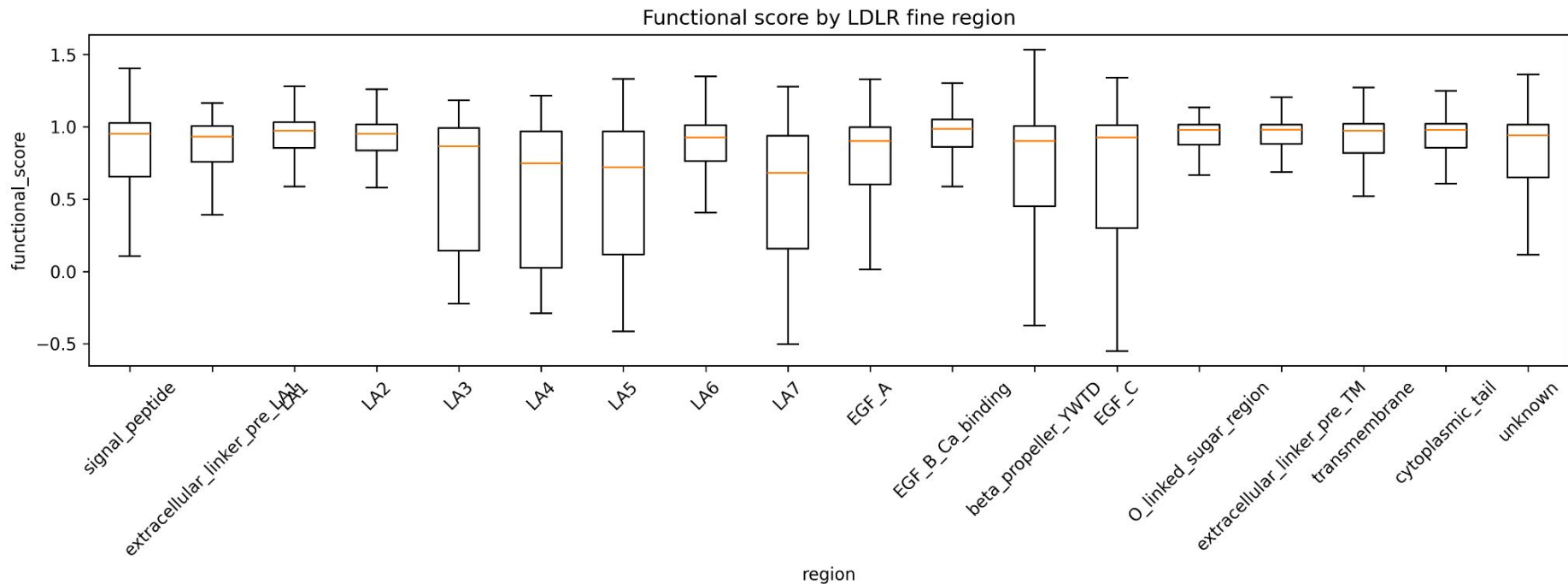


LA3–LA5 show higher cluster 3 fraction: many variants have low function but relatively preserved abundance

Part 1: Abundance Scores by Region



Part 1: Functional Scores by Region



Mutation type

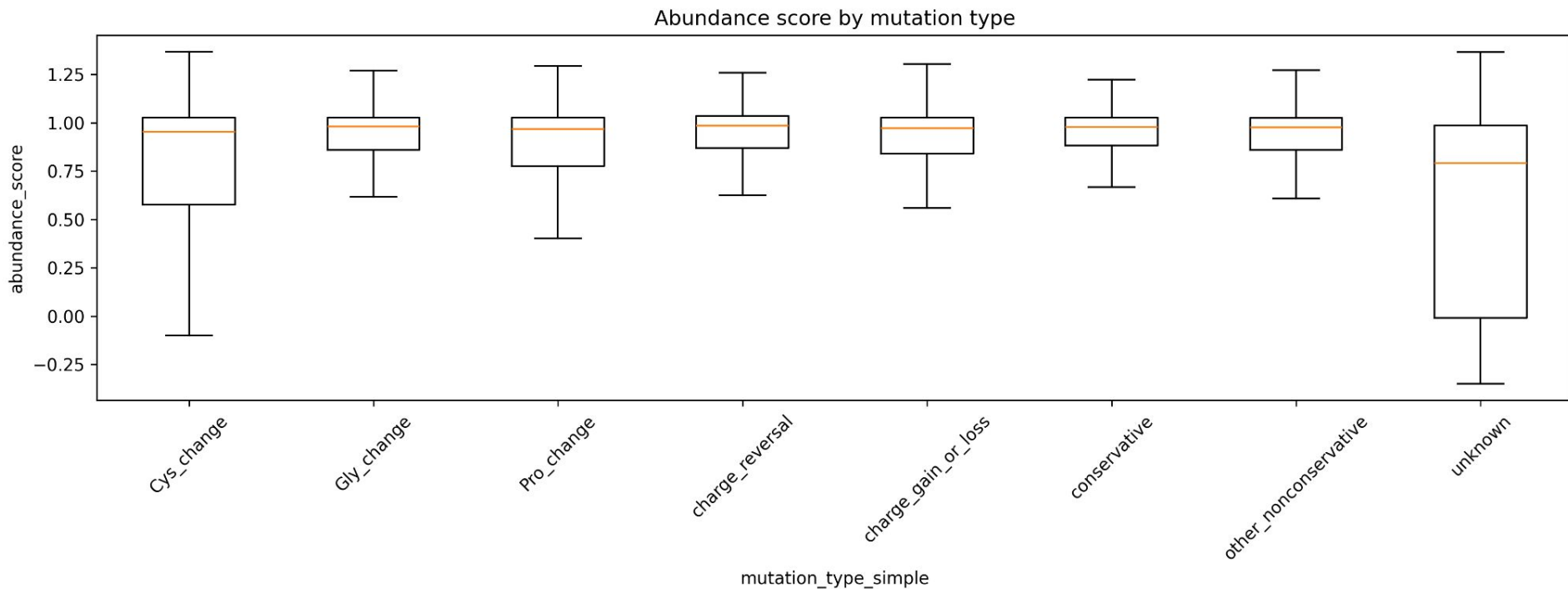
1. Conservative mutation (e.g. Leu → Ile, Asp → Glu, Lys → Arg)
2. Charge reversal (e.g. Lys → Glu, Asp → Arg)
3. Charge gain or loss (e.g. Ala → Lys, Asp → Asn)
4. Cys change (e.g. Cys → Tyr, Arg → Cys)
5. Gly change (e.g. Gly → Leu, Ser → Gly)
6. Pro change (e.g. Pro → Leu, Ala → Pro)
7. Other nonconservative mutation (e.g. Ala → Trp, Leu → Ser)
8. Unknown: synonymous, deletion, stop-gain (e.g. Gly2=, Gly5del, Trp23Ter, Arg100*)

Part 2: Cluster Fraction by Mutation Type

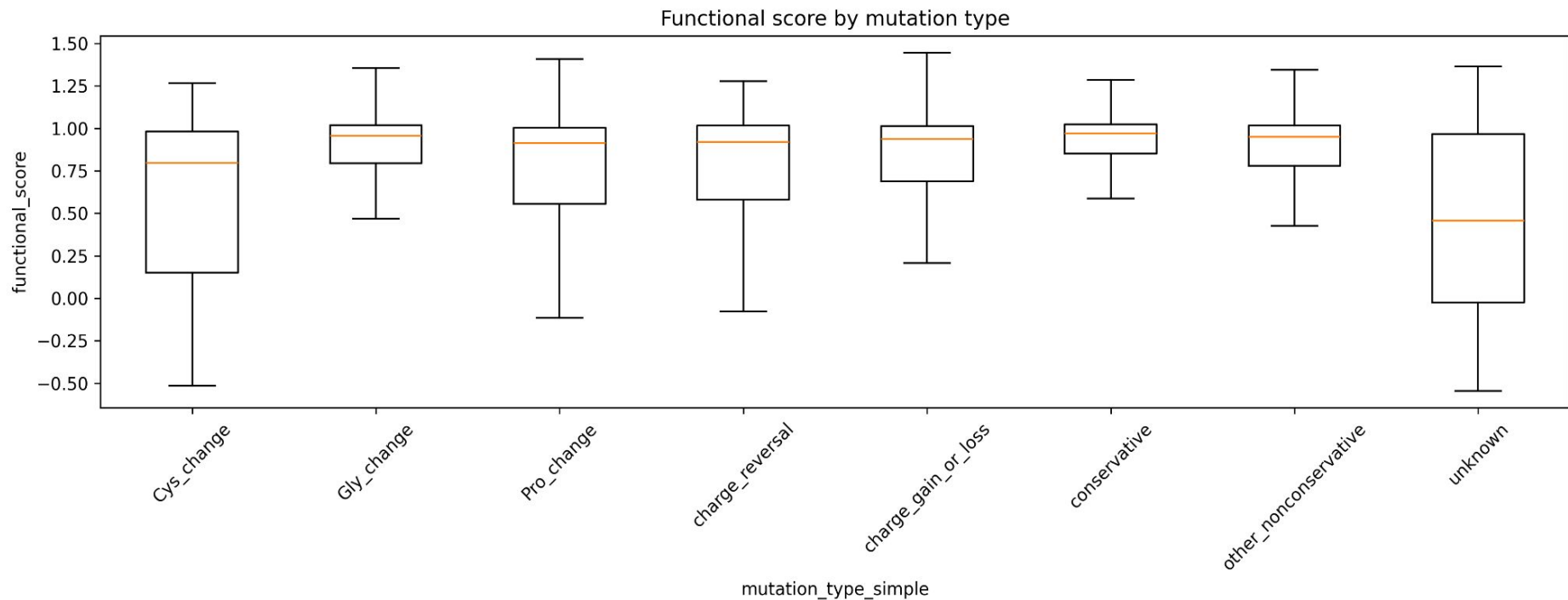


Conservative mutations are mostly cluster 0. Cys changes and charge-reversal mutations are damaging

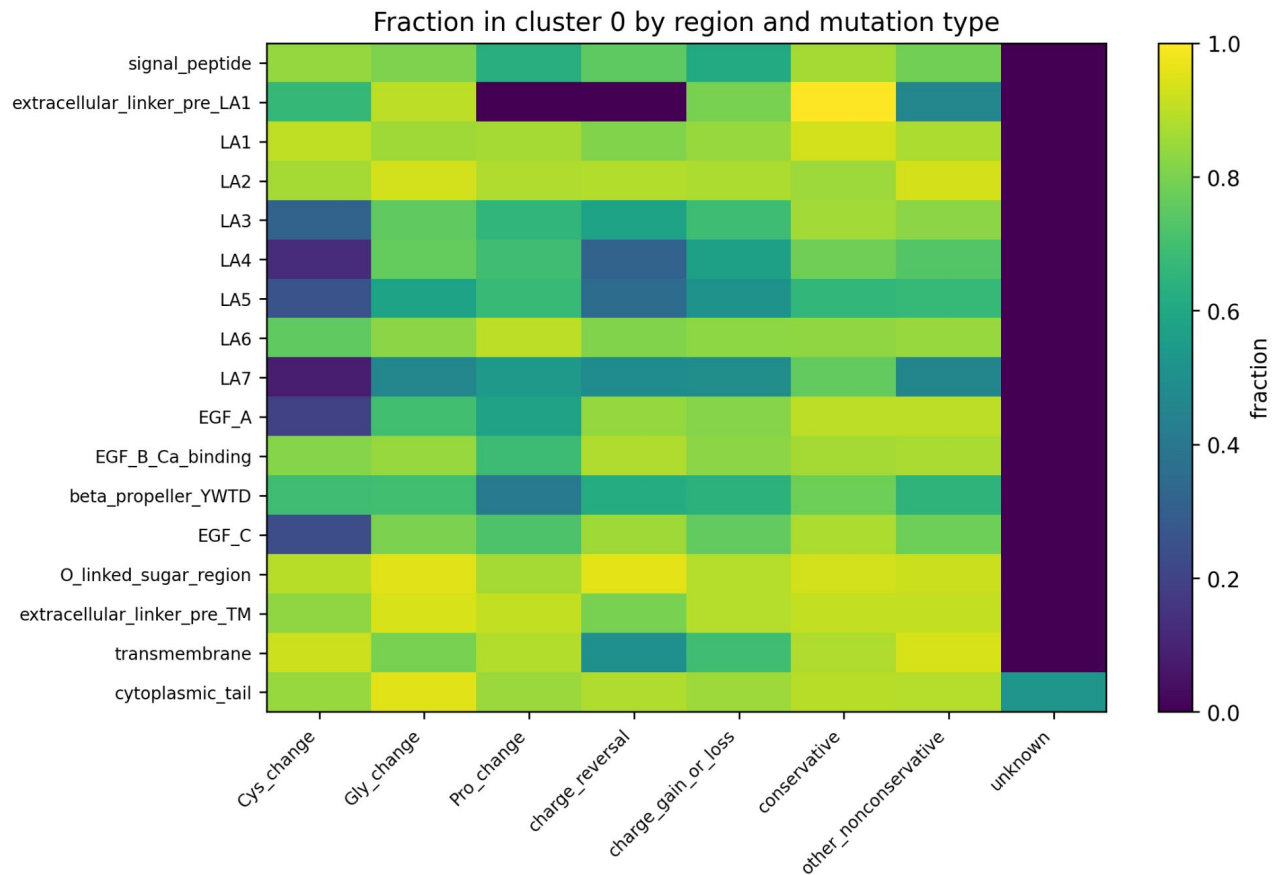
Part 2: Abundance Scores by Mutation Type



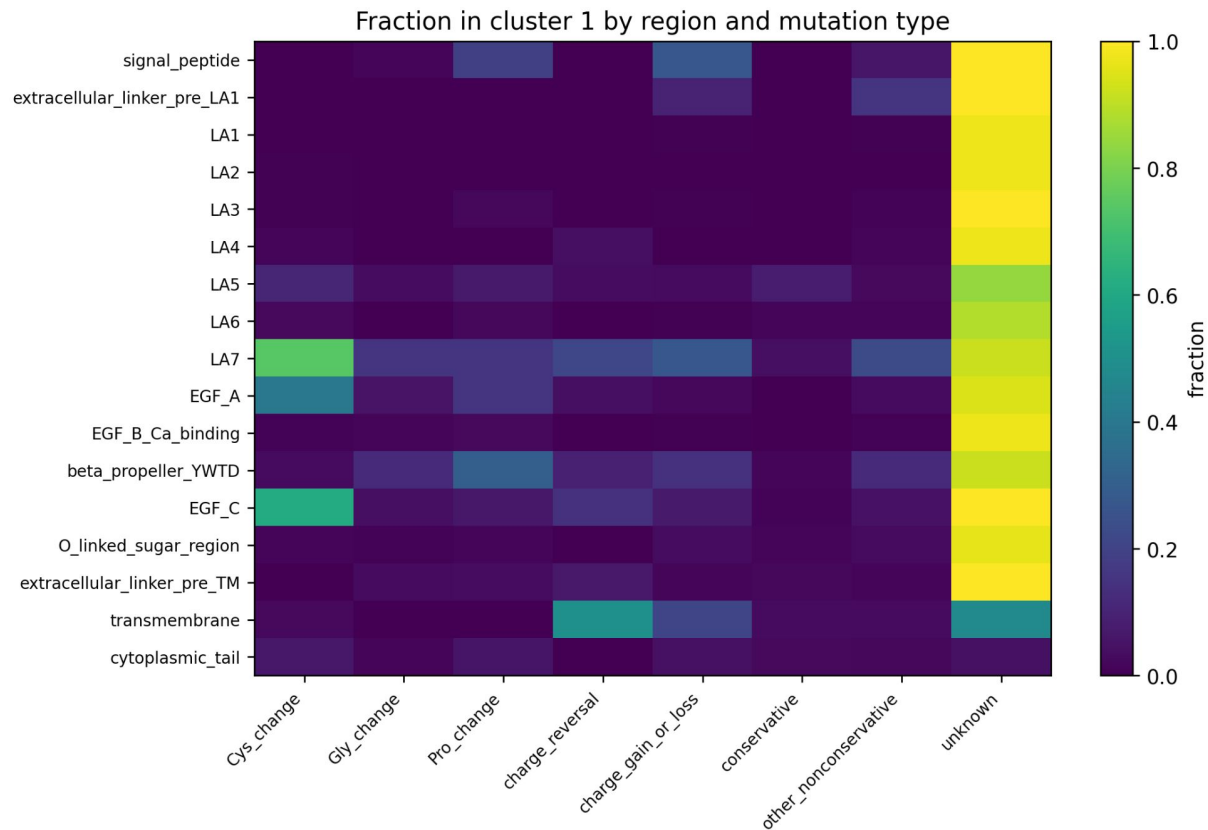
Part 2: Functional Scores by Mutation Type



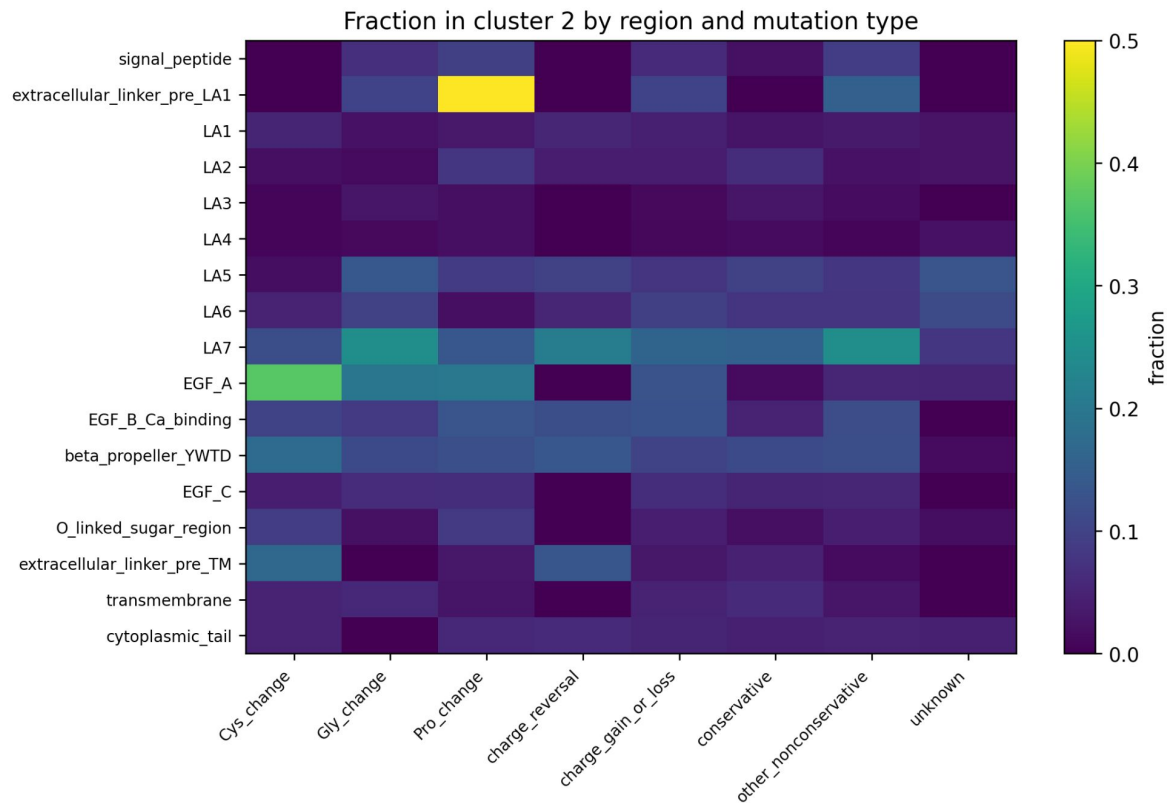
Part 3: Region × Mutation Type Heatmaps: Cluster 0



Part 3: Region × Mutation Type Heatmaps: Cluster 1



Part 3: Region × Mutation Type Heatmaps: Cluster 2



Part 3: Region × Mutation Type Heatmaps: Cluster 3

